

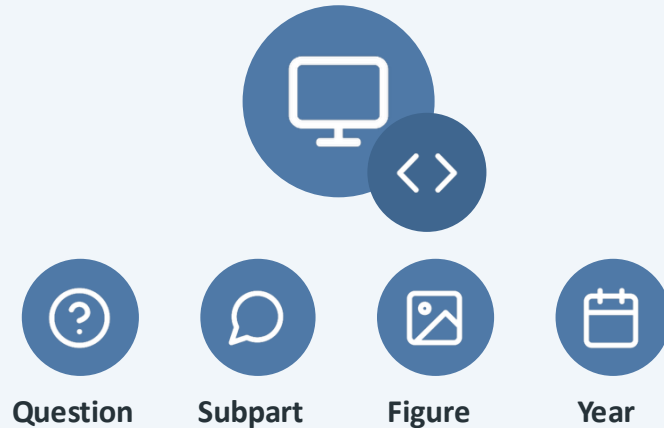
1. Sources



**USAAAO
2017–2026**

*Publicly available olympiad-style
astronomy materials*

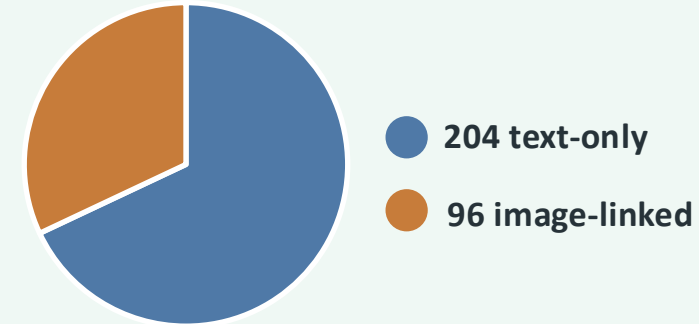
2. JSONL Conversion



Question Subpart Figure Year

*Structured conversion: question text,
solutions, subpart context, figures*

3. Benchmark Splits



*Evaluated: 300 free-responses
Reserved: 301 MCQ*

4. Model Groups



Open-weight general



Astro-specialized



Open-weight multimodal



API-served general

*Grouped by access mode, modality,
and astronomy specialization*



5. Answer Generation



Full Set



Text-only



Fixed Config

*Evaluated on full benchmark or
fair text-only subset*

6. Scoring (Automatic)



BLEU



ROUGE-L



BERTScore



Cosine Sim.



LLM-as-Judge (Fixed)

*Reference-based metrics plus a primary
GPT-5 judge protocol*

7. Analysis



Leaderboard



Metric
Disagreement



Robustness

*Leaderboard, metric-disagreement, Claude
second-judge check, and year / length /
modality robustness*

8. Main Takeaway

**Frontier models set a strong baseline.
Specialized models must justify:**



Accuracy



Cost



Reproducibility



Control

*Does astronomy specialization still help
under modern general-purpose baselines?*